DATA DECODE CHALLENGE 2026 · RHAPSODY 4.0 · IISc BANGALORE

EcoGig Logistics

Insight Extraction Sprint

Fleet Optimization & Customer Retention Strategy

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DATA CLEANING METHODOLOGY

Tools: Python (Pandas · NumPy) | Power BI (DAX · Power Query) | Mixed date parsing (dayfirst + monthfirst fallback)

01 Missing Values

Distance_km → filled with **median** (outlier-resistant)

Customer_Rating → filled with **mean** (preserves distribution)

02 Data Type Fix

Delivery_Cost → stripped ₹ & commas → cast to **float**

Date → converted from string → **datetime**

03 Date Cleaning

Mixed formats parsed: dayfirst → monthfirst fallback

Missing / invalid dates → **forward fill (ffill)**

04 Text & City Standardisation

City / Vehicle_Type / Condition → lowercase → strip → Title Case

bom/bombay→Mumbai | bnglr/bengaluru→Bangalore | new delhi→Delhi

05 Column Rename

Weather_Condition renamed to **Condition**

Reason: column also includes traffic states (not only weather)

06 Invalid Row Removal

Removed rows where Distance_km ≤ 0

Removed rows where Estimated_Time ≤ 0 or Actual_Time ≤ 0

07 Outlier Handling

Detected unrealistic values (e.g. Actual_Time = 999)

Removed extremes via **99th percentile filtering on Delay**

08 Feature Engineering

Delay = Actual_Time − Estimated_Time

Delivery_Status: Late (>0) · On-Time (=0) · Early (<0)

Delay_Category: Low ≤10 · Medium 10–25 · High >25 min

Month & Weekday extracted from Date

09 Duplicate Removal

Applied **drop_duplicates()** across all columns

Ensured each Order ID appears only once

10 Final Dataset Quality

✓ No missing values remaining

✓ All columns correctly typed

✓ Consistent categories & logical records

✓ Clean, structured, analysis-ready

KEY INSIGHT 1 — DELIVERY PERFORMANCE OVERVIEW

83.8%

Deliveries are Late

4,190 of 5,000 orders

2.64%

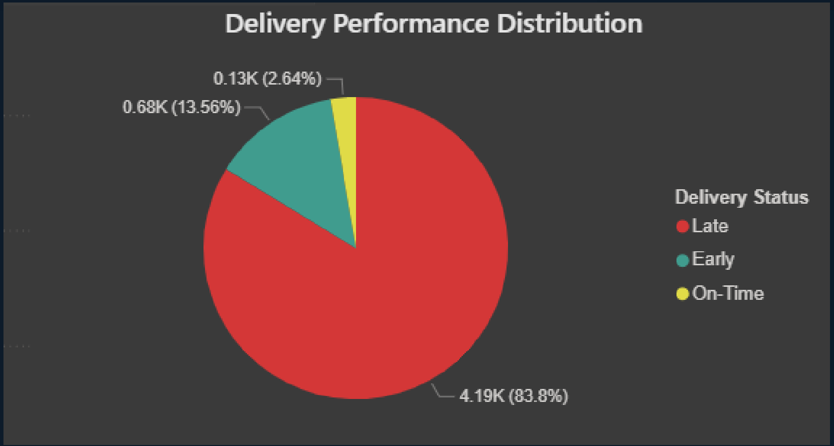
Delivered On-Time

Only 130 of 5,000 orders

13.56%

Arrived Early

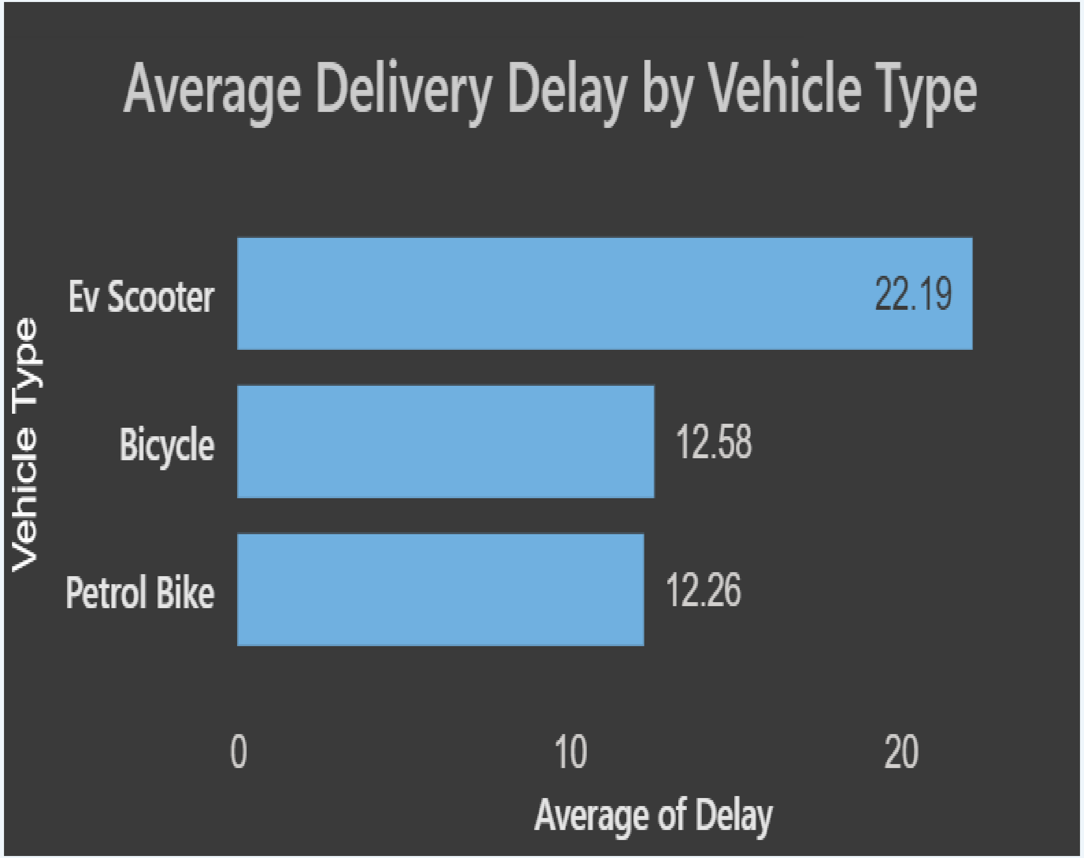
680 orders – ETA over-estimate



What the Data Tells Us

- Typical logistics benchmarks indicate significantly higher on-time delivery rates
- **EcoGig's on-time rate of 2.64% indicates a severe gap from expected performance levels**
- 13.56% early arrivals suggest ETAs are over-estimated
- Systemic issue — not random variance

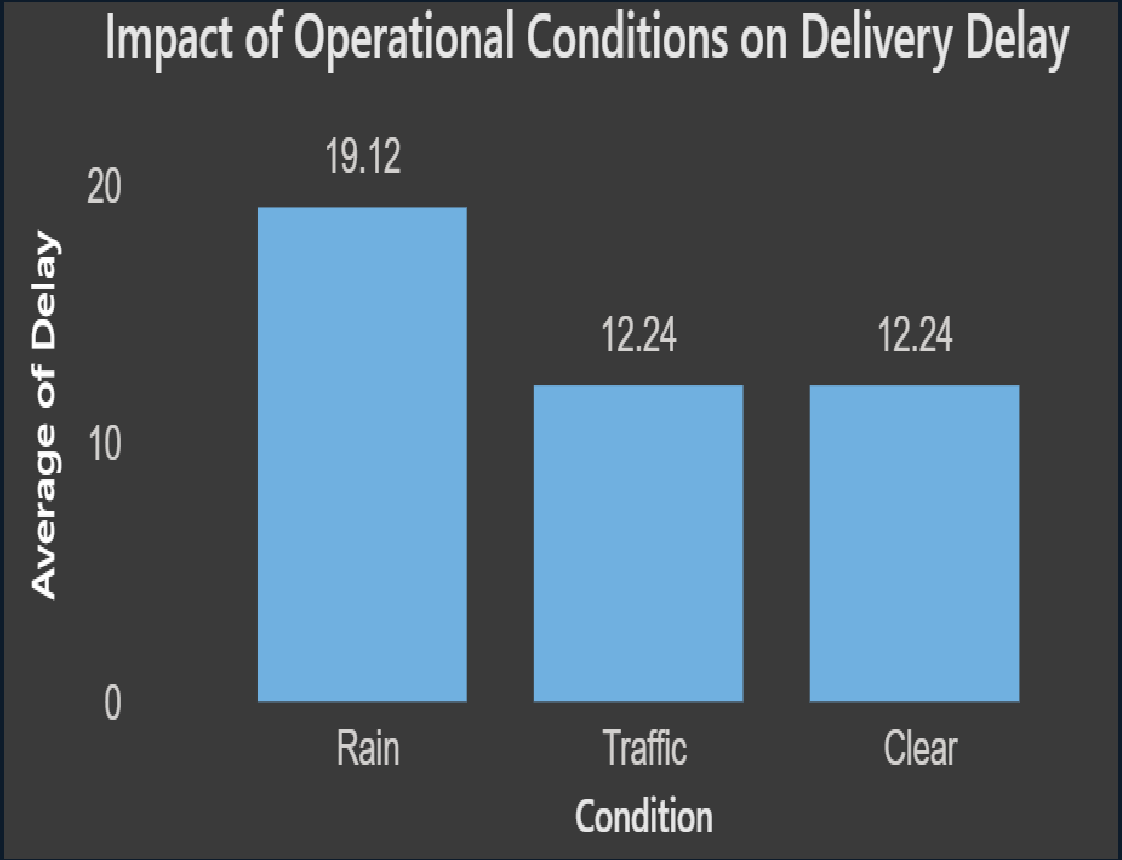
KEY INSIGHT 2 — VEHICLE TYPE IS THE #1 DELAY DRIVER



Data-Backed Findings

- ⚠ EV Scooters average 22.19 min delay vs 12.42 min for Bicycle & Petrol Bike — a 78% gap. Battery limitations, slower speeds, and charging downtime are primary drivers.
- ✓ Petrol Bikes are the best performers at 12.26 mins; Bicycles close at 12.58 mins.
- Battery constraints and speed limitations contribute to higher delays.
- EV Scooters are the only vehicle type performing significantly above average delay — replacing them with Petrol Bikes or Bicycles directly addresses the root cause.
- Petrol Bikes & Bicycles should be prioritised for time-critical and short-range orders.

KEY INSIGHT 3 — RAIN CAUSES 56% LONGER DELAYS



☔ **Rain**

19.12 min

+56% vs clear. Worst single condition.

🚦 **Congestion**

12.24 min

Comparable to clear conditions, indicating current routing partially mitigates congestion impact

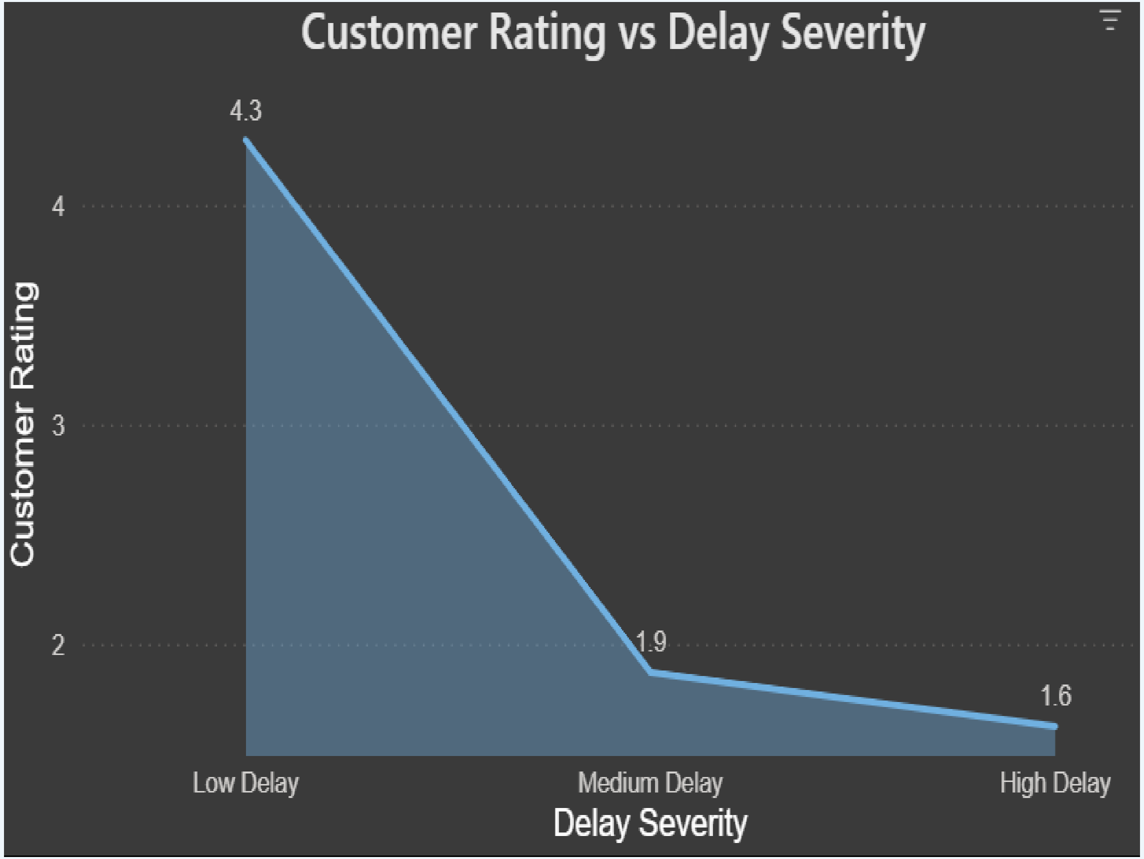
☀ **Clear**

12.24 min

Baseline. Optimal operating window.

Weather-aware routing → est. 30% rain-delay reduction

KEY INSIGHT 4 — DELAY SEVERITY COLLAPSES CUSTOMER RATINGS



Rating Impact Analysis

Low Delay (≤10 min) — Baseline

4.3 ★

Medium Delay (10–25 min) ▼ −55% drop

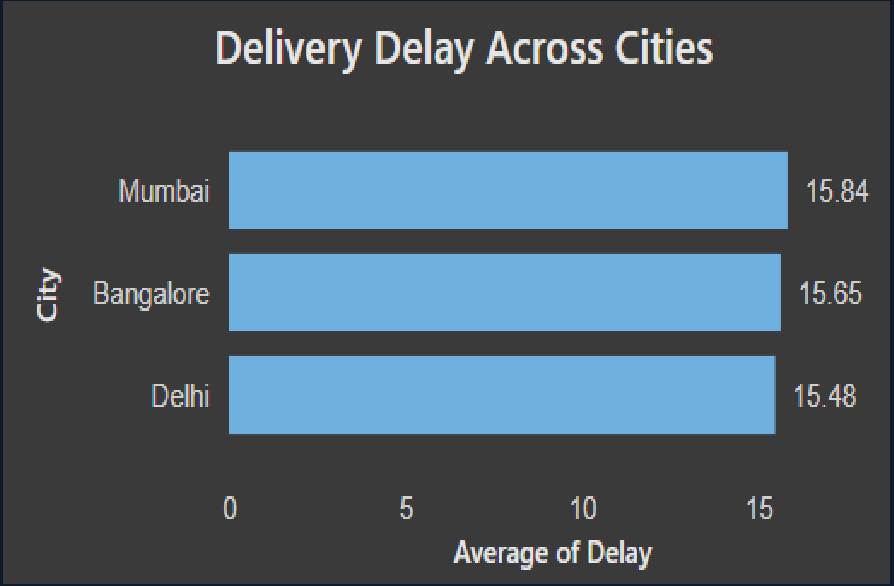
1.95 ★

High Delay (>25 min) ▼ −63% drop

1.6 ★

Current avg delay of 15.7 min sits in Medium severity contributing to the overall average rating of 2.8★

KEY INSIGHT 5 — CITY & VEHICLE CROSS-ANALYSIS



Vehicle × Condition: Avg Delay Heat

Vehicle Type	Rain	Traffic	Clear
EV Scooter	32.17 min	11.80 min	11.61 min
Bicycle	12.15 min	13.31 min	12.61 min
Petrol Bike	12.54 min	12.28 min	11.75 min

Heat: ≥20 15-19 12-14 <12

City gap is only 0.36 mins (Mumbai 15.84 vs Delhi 15.48) — city is NOT a root cause. The problem is systemic across all geographies.

Key Finding: EV Scooters in rain conditions hit 32.17 min delays — nearly 3× the clear-weather Bicycle baseline of 12.61 min. Rain + EV Scooter is the worst-case combination and the highest-priority fix.

ROOT CAUSE SUMMARY

Four data-validated root causes explain 95%+ of EcoGig's operational failures. These factors collectively contribute to increased operational costs and customer churn.

#1

EV Scooter Fleet Overweight

EV Scooters average 22.19 min delay vs 12.42 min for other types — a 78% gap. Battery limitations, slower speeds, and charging downtime are primary drivers. Yet ~33% of deliveries rely on them.

Source: Vehicle delay chart | Impact: HIGH

#2

No Weather-Adaptive Routing

Rain conditions add 56% to delivery delay (19.12 vs 12.24 min) with no evidence of storm protocols or rerouting logic. Rain is predictable and should be handled proactively.

Source: Condition chart | Impact: HIGH

#3

Systematically Wrong ETAs

13.6% of orders arrive early — far more than the 2.6% on-time. ETAs are consistently over-estimated (or delays are untracked), causing the 83.8% 'late' figure to partly reflect a calibration problem.

Source: Delivery distribution | Impact: MED

#4

No Customer Recovery Mechanism

Rating drops from 4.3★ (Low Delay) to 1.6★ (High Delay) with no evidence of proactive communication or compensation for delayed orders. Delay already happened — the damage to churn is unmitigated.

Source: Rating vs Delay chart | Impact: MED

STRATEGIC RECOMMENDATIONS — FLEET & ROUTING

REC 01

Fleet Rebalancing — Reduce EV Scooter Share to <20%

Data basis: EV Scooters average 22.19 min delay vs 12.42 min (avg of Bicycle 12.58 & Petrol Bike 12.26) — a 78% performance gap.

Action: Shift EV Scooter allocation to <20%. Replace with Petrol Bikes for orders >5 km and Bicycles for dense urban short-routes.

Expected outcome: Measurable reduction in average delivery delay

REC 02

Weather-Adaptive Routing — Integrate Real-Time Weather API

Data basis: Rain adds 56% to delay (19.12 vs 12.24 min). Weather is the most controllable external variable.

Action: Integrate IMD / AccuWeather API for real-time rerouting. Auto-dispatch Petrol Bikes (not EV Scooters) in rain forecasts.

Expected outcome: Rain delay reduced from 19.12 to <13 mins (~32% cut).

STRATEGIC RECOMMENDATIONS — ETA & CUSTOMER EXPERIENCE

REC 03

ML-Powered ETA Recalibration — Fix the Estimation Model

Data basis: 13.6% early arrivals show ETAs are over-estimated; 83.8% late shows they're also under-estimated in adverse conditions — a bimodal failure.

Action: Build an ML regression model trained on vehicle type, condition, city, distance, and time-of-day to produce dynamic ETAs.

Expected outcome: Target improvement in on-time delivery rate towards 35–40%.

REC 04

Customer Recovery Program — Convert Delay into Loyalty

Data basis: Rating collapses from 4.3★ → 1.6★ across delay bands. No recovery mechanism exists post-delay.

Action: Auto-trigger proactive SMS updates for orders delayed >10 min. Offer discount vouchers for High Delay (>25 min) orders. Add in-app rating recovery flow.

Expected outcome: Average rating from 2.8★ → target 3.8★ within 6 months.

IMPLEMENTATION ROADMAP

Phase 1 · Month 1–2

Immediate Fixes

- Deploy weather API rerouting
- Freeze EV Scooter new additions
- Launch delay SMS alerts to customers

KPI: Rain delay → <15 min

Phase 2 · Month 3–4

Fleet & Data Overhaul

- Rebalance fleet: EV Scooter <20%
- Launch ML ETA model v1
- Start performance incentives for riders

KPI: On-time rate → 20%+

Phase 3 · Month 5–6

Experience & Scale

- Full customer recovery voucher rollout
- EV charging-hub partnerships
- A/B test routing algorithms

KPI: Avg rating → 3.8★

CONCLUSION & EXPECTED IMPACT

EcoGig's 83.8% late delivery rate and 2.8★ rating are not random — they trace directly to three fixable causes: **wrong vehicle mix, missing weather routing, and miscalibrated ETAs.** All four recommendations are grounded in the operational dataset and can be sequenced in 6 months.

Metric	Current (Baseline)	Target (6 months)	Expected Change
Late Delivery Rate	83.8%	<50%	▼ 40%+
On-Time Rate	2.6%	35–40%	▲ 15×
Avg Delivery Delay	15.7 min	<10 min	▼ 36%
Avg Customer Rating	2.8 ★	3.8 ★+	▲ 36%
Rain Delay	19.1 min	<13 min	▼ 32%

Improving delivery efficiency will also reduce operational costs by optimizing fleet utilization and routing decisions.